'Use selected curve for extrusion' option and click Extrude. Press ***[Ctrl] + [A]*** to access the Attribute Editor and use its controls to edit the extrusion.

Use the Twist option to rotate the extruded polygons, and 'Taper/Taper Curve' to scale them along the curve. To control the taper exactly, open the Taper Curve section and use the graph control to set scaling along the length of the curve.

Nurbs at render time

To get rid of jagged edges of nurbs at render time, go to Attribute Editor and drop-down Tessellation. Check the Display Tessellation box and increase the 'Curvature Tolerance' by using the drop down menu. You can Increase the U and V divisions factor by typing in a higher value. This increases the meshcount when rendering, thereby outputing a smoother geometry.

Animation

Reordering inputs

The way the deformers or input nodes are assigned affect the object to a considerable extent. To reorder these, right-click on Object and go to ***Inputs > All inputs***, click the middle mouse button on the input node and place it below or above another in the same stack.

Evaluate nodes

This feature is present in the Modify menu. It's great, especially when making a tiny change in a completely rigged character setup. Just put the respective function on, make your changes, and go back to Evaluate All.

Smoothing weights

It's a bit unnerving to keep rotating the joints, or moving the Inverse Kinematics (IK) to check the influence of the joints and their weights on your skinned character.

Instead, simply animate the joints at their extreme positions. Set one key at bind pose at frame 0 and another at another pose, at, say, frame 50. Now, compare or adjust the weights as you scrub along the timeline.

Editing smooth skin bind

To change the weighting of the CVs, other than by painting weights, go to ***Windows > General Editors > Component Editor*** and under the Smooth Skin tab, type in the values for the selected vertices.

Ghosting

This is a direct translation of the classical animation technique of flipping through a handful of cell drawings to get a feel for the timing of the action you are working on.

You can control ghosting on each object in your scene with the 'Ghost Selected' and 'Unghost Selected' options windows, or with the Ghosting Information attributes. In addition to ghosting individual objects, you can now ghost entire skeletons or object hierarchies.

The appearance of ghosts in the scene view is also improved for Maya 5. The colour of ghosts before and after the current frame or keyframe lightens as their distance from it increases. This is visible only when scene view Shading is set to Wireframe.

Move along rotation axes

Use the Move tool to move along the Rotation Axis option to orient the move manipulator to the local rotation axes of the object.

Freeze joint orientation

Maya 5 lets you freeze the local joint rotation axes to match world space. To do so, go to ***Modify > Freeze Transformation*** and turn on the Orient option. If this option is turned off, the local joint rotation axes are not affected by Freeze Transformation.

Parent constraints

With a parent constraint, you can relate the position—translation and rotation—of one object to another, so that the objects behave as if they are a part of a parent-child relationship that has multiple target parents.

An object's movement can also be constrained by the average position of multiple objects.

Maintain offset

Use 'Maintain Offset' to preserve the original, relative translation, rotation, and scaling of a constrained object. This option is present in all constraint option menus

IK/FK blending lets you apply keyframe animation to joints, and also control them with IK animation. In addition to blending IK and Forward Kinematics (FK) animation over multiple frames, a blend can occur over a single frame as well. This switches IK to FK or FK to IK instantly.

Indirect skinning

Here's how you can conform the shape of the skeleton through a lattice: Construct a shape in the form of a snake—draw a curve, create a circle, and Extrude on Path. Draw a skeleton of 10 or more joints inside the model. Create an IK handle that's the length of the skeleton. Select the skin and create a lattice with ample divisions. Select Skeleton and 'Bind Skin to the Lattice'. This lets you deform the hardest of objects convincingly.

Rendering

Pelting

Reduce the effort and time you spend on adjusting those UVs when texturising characters or models: The technique is called pelting. Take your mesh, for example, the face, duplicate it and flatten out all the CVs onto a single plane, almost as if you were to hammer them. Apply a planar map to it. Now, transfer the UVs back to the original face mesh and then open the UV editor. Isn't that something!

Paint FX

Flip Tube direction

Use this to switch the direction of the current brush between Along Path (ideal for canvas painting of plants) and Along Normal (ideal for scene painting of plants). This is available as a hotkey.

A 'Force Tube to be Along